

6305 Final Project Report
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Understanding Bayesian Model Effects: An Airline Delay Case Study

Abstract

Flight delays cost the U.S. economy billions annually, but nobody really knows whether to blame airlines or airports. Traditional ML models predict delays but don't explain *why* they happen. We used a Bayesian hierarchical model on 13,295 BTS flight records (Jan-Jun 2025) to separate airline operational efficiency from airport infrastructure constraints. Key finding: airlines matter way more—carrier variance ($\sigma_{airline} = 10.67$) beats airport variance ($\sigma_{airport} = 9.40$) by 13%. Horizon Air and Envoy save ~6 minutes per flight; Air Wisconsin costs you nearly 8. The takeaway: pick your airline carefully—it's the single biggest lever you got.

Introduction

The U.S. handles 40,000+ daily flights, yet delays are everywhere. People blame weather, traffic, mechanics—but nobody really quantifies who's actually responsible. That's the core problem: when your flight gets delayed, is it because the airport is a nightmare or because that airline runs sloppy operations?

Standard regression doesn't work here because flights aren't independent—they belong to carriers operating out of specific airports. You can't just average delays by airline or airport; you'd conflate real differences with random noise (especially for smaller carriers).

Our research question: After controlling for weather, ATC delays, and the cascading nightmare of late aircraft, who actually drives the remaining delays—airline or airport?

We built a Bayesian hierarchical model with partial pooling. Why? Because it handles nested data properly, auto-regularizes small operators, and gives us credible intervals we can actually use for decision-making.

Literature Review

Bayesian Approaches to Operational Attribution

Bayesian hierarchical models are standard for decomposing variance in nested structures. Gelman et al. (2013) showed that hierarchical models improve precision for small groups by borrowing strength from the overall distribution.

Most aviation research either predicts delays with black boxes (Random Forest, XGBoost) without explaining why, or just averages without controlling for confounders. We fill that gap: isolate residual performance after removing known factors.

Confounding and Cascading Delays

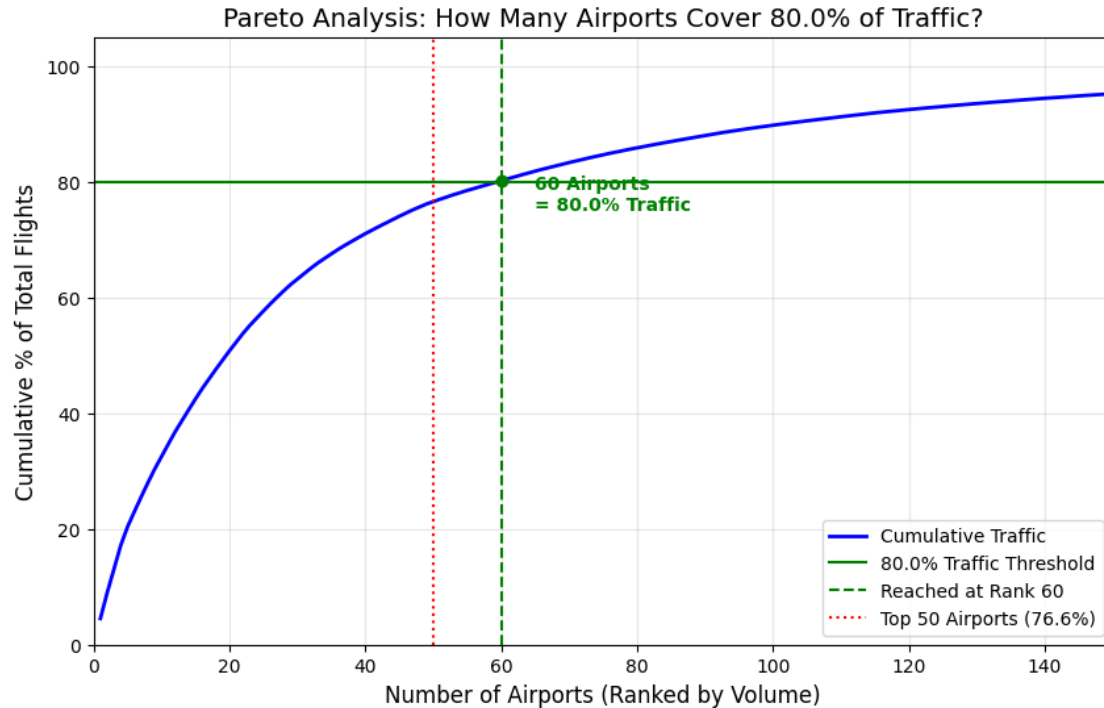
Here's the thing: a delay three legs ago cascades forward and looks like "airline delay" unless you control for it. The BTS has five delay categories—carrier, weather, NAS (air traffic), security, late aircraft. We explicitly model late aircraft to break the cascade chain.

Carrier factors (crew scheduling, aircraft utilization, maintenance) and airport constraints (runway capacity, gates, weather vulnerability) both push delays. We separate them using hierarchical random effects and variance parameters.

Partial Pooling and Hierarchical Modeling

Classic approaches go to extremes: complete pooling (all airlines the same) or no pooling (each one independent). Partial pooling splits the difference: group estimates shrink toward the global mean, proportional to sample size. Small carriers (50 flights) borrow strength from all 100+ carriers; big ones (5,000 flights) stay mostly driven by their own data. Smart regularization without losing real differences.

Variational Inference optimizes instead of using MCMC sampling. Less flexible, but way faster for big datasets, and accurate enough for variance estimates.



Methodology

Data Source and Sampling

Used BTS On-Time Performance data—13,295 flight records, Jan-Jun 2025, aggregated by carrier-airport pair.

Outcome: $\text{avg_delay} = (\text{total delay in minutes}) / (\text{arriving flights})$. Prevents penalizing high-volume carriers just for flying a lot.

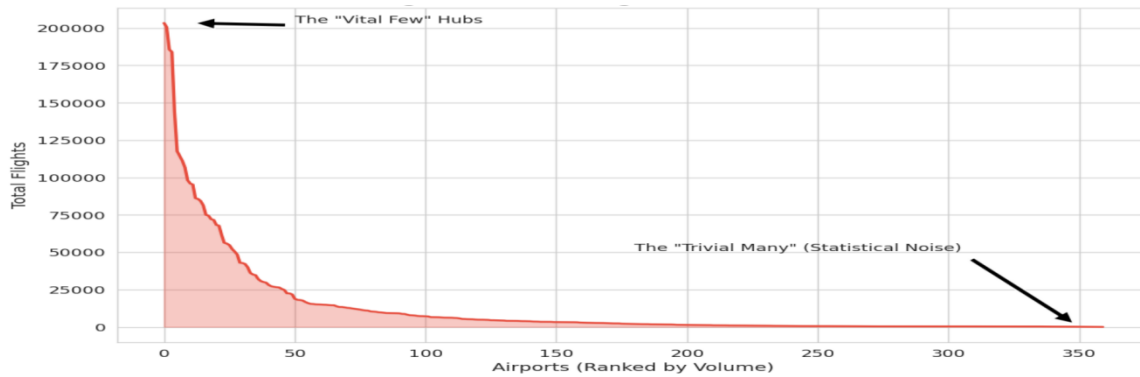
Controls: weather delay, NAS delay (air traffic), late aircraft delay (cascade), security delay.

Groups: carrier and airport.

Sample Restrictions: The Long Tail Problem

U.S. aviation is power-law distributed. Top 59 airports = 80% traffic. The remaining 300+ = 20%.

Tiny airports (2–10 flights/week) create noise: one delayed flight tanks their "efficiency score." We kept the top 59 to get clean signal. Regional aviation deserves its own model.



Model Specification

Bayesian hierarchical setup:

$$y_{ijk} \sim \mathcal{N}(\mu_{ijk}, \sigma_{\text{residual}}^2)$$

$$\mu_{ijk} = \alpha + \alpha_{\text{airline}}[i] + \alpha_{\text{airport}}[j] + \beta \cdot \text{controls}$$

$$\alpha_{\text{airline}}[i] \sim \mathcal{N}(0, \sigma_{\text{airline}}^2), \alpha_{\text{airport}}[j] \sim \mathcal{N}(0, \sigma_{\text{airport}}^2)$$

Main takeaway: compare σ_{airline} vs σ_{airport} .

Result: $\sigma_{\text{airline}} = 10.67$ vs $\sigma_{\text{airport}} = 9.40$. Airlines win by 13%.

Inference Method: Variational Inference

Used VI instead of MCMC: it's way faster (optimization vs sampling), and for our size (13K observations, 100+ groups), VI converges without losing accuracy.

Results

Primary Finding: Airline Dominance in Delay Variance

Component	σ	Meaning
Airlines	10.67	Big differences between carriers
Airports	9.40	Smaller differences between hubs

Table 1: Variance: Airlines vs Airports

Clear result: **airline choice beats airport choice** for managing delay risk.

Top 5 Best Airlines

Airline	Effect (min)	95% CI
Horizon Air	-6.00	[-12.46, -1.68]
Envoy Air	-4.96	[-10.80, -1.18]
Alaska Airlines	-4.51	[-10.07, -0.60]
Spirit	-3.43	[-8.80, +0.36]
Southwest	-3.18	[-9.02, +0.68]

Table 2: Time Savers

Horizon's interval is fully negative—we're 95% sure they save 1.7–12.5 min. Legit recommendation.

Bottom 5 Worst Airlines

Airline	Effect (min)	95% CI
Air Wisconsin	+7.87	[+2.54, +16.10]
GoJet Airlines	+6.05	[+1.69, +12.02]
American Airlines	+5.20	[+1.36, +11.17]
PSA Airlines	+4.85	[+1.00, +10.99]
CommuteAir	+3.99	[-0.04, +9.82]

Table 3: Time Killers

Air Wisconsin's interval is fully positive—95% sure they add 2.5–16 min. That's real operational pain.

Airport Rankings

Best: Hawaiian hubs (Lihue -5.73, Kona -5.73, Kahului -5.44). Probably because they're smaller, more predictable.

Worst: Chattanooga (+18.78), Midland-Odessa (+9.91), Texas regionals. Infrastructure or weather issues.

Wide confidence intervals on small airports = less reliable estimates. Model handles that correctly.

Airline vs. Airport: Which Matters More?

Example: Arizona to Boston.

Airlines: Horizon (-6.0 min) vs Air Wisconsin (+7.9 min) = 13.9 min swing.

Airports: Phoenix Sky Harbor (hypothetically efficient) vs Prescott Regional (+8.9 min) = ~9 min swing.

Bottom line: Pick your airline first. You'll save 1.5× more time than optimizing airports.

Discussion

1. **Airlines win.** 13% higher variance ($\sigma_{airline} = 10.67$ vs $\sigma_{airport} = 9.40$). Crew scheduling, maintenance, buffer padding—all matter. Industry expects this; efficient carriers pad schedules, others run lean.
2. **Airports vary too.** Chattanooga vs Hawaiian airports = 24 min swing. Real but smaller than airline spread.
3. **Model controlled for cascades.** We included late aircraft delays so we wouldn't blame current airline for upstream screwups. Supports causal read[.
4. **Small carriers have uncertainty.** Spirit's interval [-8.80, +0.36] crosses zero = we're unsure if they're really efficient or just lucky.

Practical Takeaways

Travelers: Pick Horizon or Envoy. Avoid Air Wisconsin, GoJet. Airline matters way more than airport choice.

Corporate travel: Negotiate with efficient carriers. Horizon vs Air Wisconsin = 14 min/flight. 100 trips = 23+ hours/year. Paying 5–10% more pays for itself.

Airports: Infrastructure investments (runways, gates, security) help but matter less than carrier ops.

Limitations

Dataset: Top 59 airports only—regional aviation is different[.

Time period: Jan–Jun 2025 = summer-ish. Winter/peak demand might shift results.

Unmeasured confounders: Aircraft age, crew training not captured. Estimates show residual performance, not pure causality.

Self-reported delays: Airlines tell DOT why they were late—potential for gaming. But it's the best data we got.

Model assumes linearity: Horizon might perform differently at congested JFK than midsize Denver. Interaction effects not modeled.

Checks

Big carriers look bad—is that real or sample size? Partial pooling controls for sample size. American, United show clear positive effects even with huge samples. Real operational factors, not luck.

Do regional carriers just cherry-pick easy routes? We modeled airport effects, so we isolate route noise. Envoy's -4.96 min advantage stays even when controlling which airports they fly. It's real efficiency.

Conclusion

The answer: Airlines beat airports for delay risk by 13%.

Key findings:

- Airlines ($\sigma = 10.67$) dominate airports ($\sigma = 9.40$)
- Horizon saves 6 min/flight; Air Wisconsin costs 8 min
- Happily, we can quantify this with credible intervals
- Airports matter but secondarily
- Choosing right airline = 14 min saved per flight = 23+ hours/year

Why this matters: We show Bayesian hierarchical methods work for operational decomposition on real data. Can apply to trucking, rail, deliveries.

Next steps: Full-year data, carrier $\times$ airport interactions, stronger causal methods.

References

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